A Comparative Analysis of Robust Penalized Estimators for Periodic Time Series

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Problem Statement & Objectives

- **Motivation:** Address non-stationary traffic data for robust, interpretable forecasting based on F.Rodrigues study [Rodrigues, 2023]
- **Primary Objective:**
  - Conduct a rigorous comparative analysis of LSA and MTE
  - Perform MEB procedure to estimate CI for the models and VIPs of features
  - Develop an innovative adaptive LAD-Lasso, weighted by Variable Inclusion Probabilities (VIPs) to prioritize statistically significant harmonic components.
- **Framework:** Implement methods through LAD and LS regression frameworks.
- **Model Optimization:** Utilize AIC and BIC minimization.

Data Source: TRANSFOR Competition

- **Origin:** 2013 TRANSportation Data FORecasting Competition (TRANSFOR).
- **Provider:** National Technical University of Athens.
- **Content:** Traffic volume and occupancy measurements.
- **Period:** April 2000.
- **Frequency:** Every 90 seconds. Mean aggregated to 180 seconds (3-minutes intervals)
- **Focus:** Traffic volume.

Study Area: Alexandras Avenue, Athens

- **Location:** Alexandras Avenue, a key arterial avenue in Athens, Greece.
- **Detectors:** Seven loop detectors ($L \in \{A, B, C, D, E, F, G\}$).
  - Westbound traffic: Detectors A, B, C, D.
  - Eastbound traffic: Detectors E, F, G.

Figure: *Locations of the Loop detectors on Alexandras Avenue. This image is taken from [Giacomazzo and Kamarianakis, 2020].*

Data: Characteristics & Anomalies

- **Raw Data Challenges:** Large volume, low signal-to-noise ratio.
- **Temporal Aggregation:** Data aggregated to 3-minute intervals (480 observations per day per detector).
  - Balances smoothing with high-resolution forecasting needs.
- **Observed Patterns:** Noisy data, low values at the end of each workday.
- **Notable Anomalies:**
  - April 10th: Significant anomaly after elections. (referendum: GRD vs €)
  - April 20th: Greek basketball club celebration near Alexandras Avenue.
  - Traffic incidents.
  - Weather
  - ...

Data: Plot Traffic Volume across locations

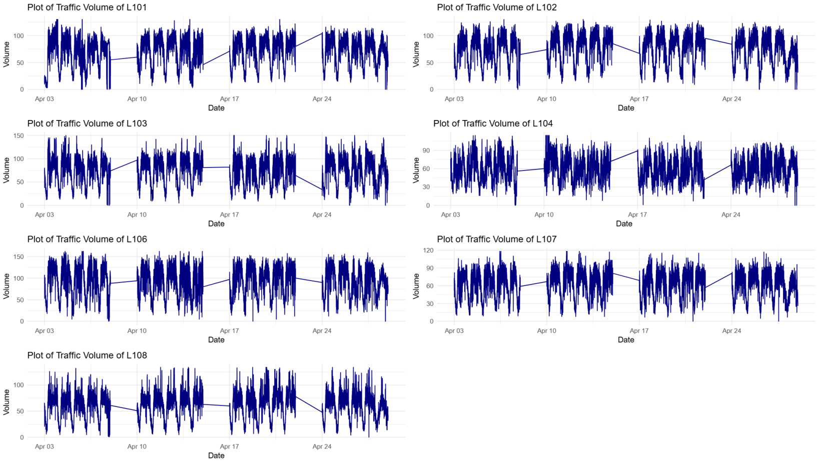

Figure: *Traffic volume data on locations of the 7 loop detectors.*

Data: Remarks

Key Observations:

- Clear seasonal effect (daily periodicity).
- Noisy data, with traffic volume ranging from 0 to 160 vehicles.
- Visual evidence singularities across all locations.
- Highlights the challenges for effective model training.
- Volume $\in [0 - 162] \Rightarrow \Delta$(Volume - Historic Averages) is bounded

Methods: Harmonic Seasonal Model (HSM)

- **Model Definition:**

$$x_t = m_t + \sum_{i=1}^{\lfloor T/2 \rfloor} s_i \sin\left(\frac{2\pi i t}{T}\right) + c_i \cos\left(\frac{2\pi i t}{T}\right) + z_t$$

  - m_t: Trend component.
  - s_i, c_i: Unknown parameters to be estimated
  - $\lfloor T/2 \rfloor$: Upper bound representing possible cycles. In our study $T = 2400$.

- **Coefficient Estimation:**
  - Median Regression (l_1 norm): Robust to outliers, non-smooth optimization function.
  - Mean Regression (l_2 norm): Sensitive to outliers, smooth, but has lots of limitations (f.i. multicollinearity, heteroskedasticity, autocorrrelation, etc)

Methods: Harmonic Seasonal Model (HSM) - Practical Application

- **Empirical Evidence:** Low harmonic frequencies improve short-term traffic forecasting accuracy.
- **Scope of Analysis:** Focus on the first 500 harmonic frequencies (out of 2400).
- **Benefits:**
  - Reduces model complexity.
  - Provides a computationally inexpensive and easy-to-implement benchmark.
- **Future Steps:** Explore more complex models to capture seasonality and assess their out-performance.

Methods: Adaptive LAD Lasso - Introduction

- **Adaptive LASSO [Zou, 2006]:** A two-step procedure designed to achieve "oracle properties."
  - First step: Auxiliary LASSO regression.
  - Second step: Apply LASSO with adjusted penalty parameter based on non-zero coefficients from the first step.

$$\hat{\beta}^{(n)} = \arg\min_{\beta} \left\| \mathbf{y} - \sum_{j=1}^{p} x_j \beta_j \right\| + \lambda_n \sum_{j=1}^{p} |\beta_j|,$$

converges in probability to the minimizer of the function

$$\mathcal{L}_1(\beta) = (\beta - \beta^*)^T \left(\frac{1}{n} X^\top X \right) (u - \beta^*) + \lambda_0 \sum_{j=1}^{p} |\beta_j|,$$

where λ_n is the parameter of LASSO estimation

Methods: Adaptive LAD Lasso - Properties & Implementation

Oracle Property:
Assume that:

$$\begin{cases} \lambda_n/\sqrt{n} \to 0, \\ \lambda_n n^{(\gamma-1)/2} \to \infty, \gamma > 0, \\ \hat{\boldsymbol{\Sigma}} \approx \frac{1}{n} X^\top X = \begin{pmatrix} \hat{\Sigma}_{11} & \hat{\Sigma}_{12} \\ \hat{\Sigma}_{21} & \hat{\Sigma}_{22} \end{pmatrix}, \text{ where } \hat{\Sigma}_{11} \text{ is a } p \times p \text{ matrix,} \end{cases}$$

then

- ☞ Consistency in variable selection: $\lim_{n\to\infty} P(\mathcal{A}_n^* = \mathcal{A}) = 1$.
- ☞ Asymptotic normality: $\sqrt{n}(\hat{\beta}_A^{(n)} - \beta_A^*) \to_d \mathcal{N}(0, \sigma^2 \times \hat{\Sigma}_{11}^{-1})$.

- **Key Properties:**
  - Consistency in variable selection: $\lim_{n\to\infty} P(\mathcal{A}_n^* = \mathcal{A}) = 1$.
  - Asymptotic normality: $\sqrt{n}(\hat{\beta}_A^{(n)} - \beta_A^*) \to_d \mathcal{N}(0, \sigma^2 \times \hat{\Sigma}_{11}^{-1})$.
  - Oracle properties closely tied to super-efficiency phenomenon [Lehman and Cassela, 1998].

Methods: Least Squares Approximation (LSA) of Unified LASSO - Overview

- **Core Idea:** Integrates LARS and LASSO estimation techniques in several scenarios [Wang and Leng, 2007]. We focus on the latter part.

$$
\begin{aligned}
\arg\min_{\beta\in\mathbb{R}^d}\left(\frac{\mathcal{L}_n(\beta)}{n}\right) &= n^{-1}\mathcal{L}_n(\tilde{\beta}) + n^{-1}\cancelto{0}{\dot{\mathcal{L}}_n(\tilde{\beta})^\top}(\beta-\tilde{\beta}) + \frac{1}{2}(\beta-\tilde{\beta})^\top\left\{\frac{1}{n}\ddot{\mathcal{L}}_n(\tilde{\beta})\right\}(\beta-\tilde{\beta}) \\
&= \arg\min_{\beta\in\mathbb{R}^d}\left(\frac{\mathcal{L}_n(\tilde{\beta})}{n} + (\beta-\tilde{\beta})^\top\frac{\ddot{\mathcal{L}}_n(\tilde{\beta})}{n}(\beta-\tilde{\beta})\right) \qquad (*) \\
&\approx \arg\min_{\beta\in\mathbb{R}^d}\left((\beta-\tilde{\beta})^\top\hat{\Sigma}^{-1}(\beta-\tilde{\beta})\right)
\end{aligned}
$$

where $\mathcal{L}_n$ is the loss function of an unpenalized estimator, $\tilde{\beta}$ is its minimizer and $\mathbb{E}\left[\frac{\ddot{\mathcal{L}}_n(\tilde{\beta})}{n}\right] \overset{n\to\infty}{\to} \hat{\Sigma}^{-1}$.

Key Note: Simplifies LASSO application by reformulating the objective into an asymptotically equivalent least squares problem.

Methods: Least Squares Approximation (LSA) - Mathematical Basis

- **Covariance Matrix Estimation for LS:** $cov(x^S)$.
- **Covariance Matrix Estimation for LAD:** Leverages Pollard's [Pollard, 1991] asymptotic covariance matrix.

$$\hat{\Sigma}^{-1} \approx 4 f_e^2(0)\mathrm{cov}(x^S)$$

- **Advantages:**
  - When combined with Taylor approximation of adaptive LASSO penalty and BIC-type tuning, LSA estimator can be as efficient as the oracle.
  - Outperforms traditional methods, robust to outlier effects.
  - Manages non-smooth loss functions (e.g., LAD).
- **Applicability:** Can perform conventional LASSO, robust (l_1) LASSO, and their adaptive forms if the sample covariance matrix is non-singular.

Methods: Least Squares Approximation (LSA) - Properties & Tuning

- **Proven Properties:**
  - $\sqrt{n}$ consistent if regularization diminishes to zero faster than $\sqrt{n}$.
  - Consistently captures the true model (truly zero coefficients estimated as zero).
  - Achieves oracle efficiency.
- **Computational Efficiency:** Minimizes BIC criterion using a rule of thumb for $\lambda_j = \frac{\lambda_0}{|\hat{\beta}_j|^\gamma}$ to avoid exhaustive search.
- **Implementation:** Implemented in R software, available from the authors (lars.lsa function).

Methods: Penalized Maximum Tangent Likelihood Estimation (MTE) - Overview

- **Motivation:** More reliable alternative to LSA, especially in the presence of outliers [Qin et al., 2017].
- **Key Features:**
  - $\sqrt{n}$ consistent and possesses the oracle property.
  - Includes several well-known methods as special instances.
  - Achieves an ideal rate of convergence ($\frac{\sqrt{\ln(d)}}{n}$), outperforming existing methods like Lasso.

Methods: Penalized Maximum Tangent Likelihood Estimation (MTE) - Mathematical Basis

- **Core Principle:** Based on maximizing the tangent likelihood.

$$\tilde{\beta} = \arg\max_{\beta \in \mathbb{R}^d} \left(\underbrace{\sum_{i \in \mathcal{A}} ln(f(y_i - x_i^T \beta))}_{\text{minimizing KL divergence}} - \underbrace{\frac{1}{t} \sum_{i \in \mathcal{A}} f(y_i - x_i^T \beta)}_{\text{minimizing } \ell_2 \text{ metric}} + n \sum_{j=1}^{d} \frac{\partial}{\partial \beta} p_{\lambda_j}(|\beta_j|) \right)$$

- **Asymptotic Properties:**
  - Probabilistic convergence of coefficients.
  - Central Limit Theorem under distributional convergence.
  - Consistent estimator with asymptotic normality under specific conditions.
- **Penalization Scheme:** Combines MTE with penalization for variable selection.

Methods: Penalized Maximum Tangent Likelihood Estimation (MTE) - Oracle Properties

- **Conditions for Oracle Property:** Requires specific conditions on penalty terms and coefficient vectors.
- **Sparsity and Asymptotic Normality:** Ensures that zero coefficients are estimated as zero with high probability, and non-zero coefficients are asymptotically normal.
- **Restricted Eigenvalue Condition (REC):** Ensures that the loss function contains enough curvature for accurate ℓ_2 norm estimation.
- **Computational Efficiency:** Accelerated by coordinate descent optimization algorithm.
- **Hyperparameter Tuning:** Greedy tuning of λ in a constrained domain (e.g., $(0, 0.1]$).

Methods: Maximum Entropy Bootstrap (MEBoot) - Purpose

- **Challenge:** Estimating a statistic's sampling distribution for time series data is difficult due to time dependency and non-stationarity.
- **MEBoot Solution:** A principled, nonparametric resampling strategy.
- **Key Benefits:**
  - Produces smooth, continuous-valued samples.
  - Preserves the time-order and dependency structure of the original data.
  - Generates the least biased distribution compatible with empirical constraints.
- **Suitability:** Very well suited for our objective of robust inference under uncertainty.

Methods: Maximum Entropy Bootstrap (MEBoot) - Methodology

- **Foundation:** H. Vinod extended the Wiener-Kolmogorov-Khinchine (WKK) process [Vinod and de Lacalle, 2017].
- **Modern Approach:** Computer-intensive method for constructing time series ensembles.
- **Utility:** Particularly useful for short, non-stationary time series or those with regime changes, gaps, and sudden jumps.
- **Limitations:**
  - Reliance on accurate model specification.
  - Difficulties with structural breaks or abrupt regime shifts.
- **Assumption:** Underlying process does not possess infinite memory (enables ergodic theorem).

Methods: Maximum Entropy Bootstrap (MEBoot) - Methodology

- This method allows us to estimate CIs for LSA and MTE
- We can estimate feature importance by calculating VIPs
- We can use VIPs as weights to a post-hoc Adaptive LAD LASSO framework.

Methods: Maximum Entropy Bootstrap (MEBoot) - Steps

1. **Ranking:** Sort observed time series data, create index vector for original time locations.
2. **Interval Boundaries:** Compute midpoints of consecutive order statistics.
3. **Extending Tails:** Define outermost intervals using trimmed mean of sequential differences.
4. **Assigning Interval Means (Maximum Entropy):** Assign means within each interval to preserve observed data mean.
5. **Sampling from ME Distribution:** Create uniform random variables and convert to sample quantiles.
6. **Restoring Order:** Rearrange sorted quantiles to match original temporal order.
7. **Repeat:** Multiple resamples for robust inference and variability assessment.

- **Application:** 20 repetitions performed to obtain Variable Inclusion Probabilities (VIPs).

Methods: Hyperparameter Tuning: AIC vs. BIC

- **Purpose:** Find the best model describing the data generating mechanism.
- **AIC (Akaike Information Criterion):** $2k - \log(L)$
  - Based on information theory.
  - Values structural accuracy over parsimony.
  - Prefers more flexible models, suitable for capturing nuanced patterns.
- **BIC (Schwarz/Bayesian Information Criterion):** $k \log(n) - \log(L)$
  - Based on Bayesian framework.
  - Imposes a higher penalty for each new parameter, especially with large datasets.
  - Prefers simpler, more parsimonious models, usually more stable.
- **Practicality:** Neither is clearly superior; often beneficial to consult both.

Methods: Metrics Used for Model Comparison

- **Comprehensive Evaluation:** Broad range of out-of-sample measures used.
- **Mean Absolute Error (MAE):**

$$\left(MAE = \frac{1}{T} \sum_{t=1}^{T} |y_t - \hat{y}_t| \right)$$

  - Average of absolute values of errors.
  - Scale-dependent accuracy metric.
- **Root Mean Square Error (RMSE):**

$$\left(RMSE = \sqrt{\frac{1}{T} \sum_{t=1}^{T} (y_t - \hat{y}_t)^2)} \right)$$

  - Square root of the mean of squared deviations.
  - Integrates magnitudes of individual prediction mistakes.
  - Scale-dependent.

Methods: Detailed Error Metrics (Cont.)

- **Huber Loss (HL):**

$$L_\delta(y,\hat{y}) = \begin{cases} \frac{1}{2}(y - f(x))^2 = \frac{1}{2}\varepsilon^2 & \text{if } |y - f(x)| \leq \delta, \\ \delta\left(|y - f(x)| - \frac{1}{2}\delta\right) = \delta\left(|\varepsilon| - \frac{1}{2}\delta\right) & \text{otherwise,} \end{cases}$$

where $\varepsilon = y - f(x)$ denotes the residuals.

  - Combines properties of MAE and RMSE.
  - Quadratic near origin, linear for large residuals.
  - Less sensitive to outliers than squared loss.

- **Symmetric Mean Absolute Percentage Error (sMAPE):**

$$SMAPE = 100 \cdot \frac{1}{n} \sum_{t=1}^{n} \frac{2|y_t - \hat{y}_t|}{|y_t| + |\hat{y}_t|}$$

- **Symmetric Median Absolute Percentage Error (SMdAPE):**

$$SMdAPE = 100 \cdot \text{Med}\left(\frac{2|y_t - \hat{y}_t|}{|y_t| + |\hat{y}_t|}\right)$$

.

Context is Key: Both of SMAPE or SMdAPE depend on data characteristics and application (proposed in [Makridakis et al., 2022]).

Methods: Overall Methodology Pipeline

- **Step 1: Data Resampling:** Original dataset resampled using Maximum Entropy Bootstrap (20 synthetic datasets).
- **Step 2: Harmonic Seasonal Model (HSM):** Applied to each synthetic dataset (using ℓ_1 or ℓ_2 norm).
- **Step 3: Robust Estimator Application:**
  - Least Squares Approximation (LSA) with AIC/BIC minimization.
  - Penalized Maximum Tangent Likelihood Estimation (MTE) with greedy hyperparameter tuning.
- **Step 4: Variable Inclusion Probabilities (VIPs):** LSA and MTE outputs used to construct VIPs.
- **Step 5: Adaptive ℓ_1-Lasso:** VIPs inform the weighting scheme for the final Adaptive ℓ_1-Lasso regression.
- **Step 6: Cross-Validation:** 10-fold (Bi-daily) Leave-One-Block-Out Cross-Validation (LOBOCV) for each model greedily searching for $\lambda \in \mathcal{D} = [0.02, 0.1]$, with step size 0.02.

Methods: Pipeline Illustration

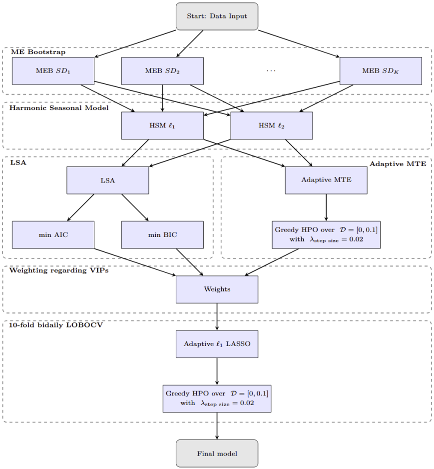

Figure 3: *The flowchart of our comparative analysis. We first generate the MEBoot datasets we apply Harmonic Seasonal Model using ℓ_1 and ℓ_2 norm. Afterwards, we apply the models we want to compare: LSA and MTE for ℓ_1 and ℓ_2 norm minimization. In the MTE scenario we apply a greedy fine tuning algorithm for λ in a set of $\mathcal{D} = [0, 0.1]$ with step values to be 0.02. On the other hand, we leverage the minimization of AIC and BIC on the LSA based methods. These procedures are applied on each MEBoot dataset and we collect the Variable Inclusion probabilities. The VIPs are fed into the Adaptive $\ell 1$ LASSO as weights. We apply again the greedy hyper parameter tuning algorithm over the set $\mathcal{D}$ with step equal to 0.02 for λ values.*

Results Overview: Model Family & Feature Importance

- **Model Family:**
  $\{\ell_x M_i \mid x \in \{1,2\}, M \in \{\text{MTE, LSA}\}, i \in \{\text{AIC, BIC}\}\}$.
- **Feature Importance:** Quantified using VIPs from MEBoot.
- **Location Focus:** Primary analysis on L102 (B) due to anticipated consistent traffic flow.
- **Initial Observation:**
  - Consistent pattern of significant harmonic frequencies across most methods (ℓ_1/ℓ_2MTE, ℓ_2LSA).
  - Primary exception: ℓ_1-norm Least Squares Approximation (ℓ_1LSA) yields notably sparser models.

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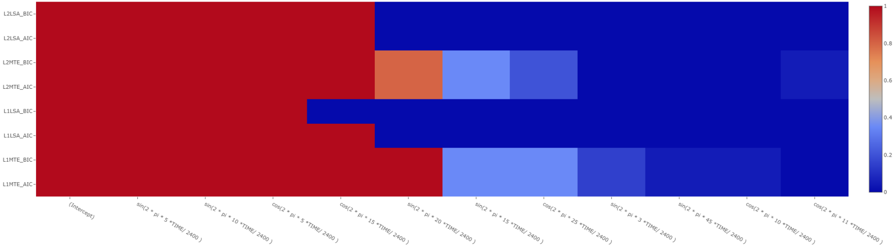


Figure: *Variable Inclusion Probabilitis across all methods for location L102.*

Results: L102 - VIPs

Features \ Methods	**L1MTE$_{AIC/BIC}$**	**L1LSA$_{AIC}$**	**L1LSA$_{BIC}$**	**L2MTE$_{AIC/BIC}$**	**L2LSA$_{AIC/BIC}$**
$(Intercept)$	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
$sin\left(2\pi \cdot 5\frac{t}{2400}\right)$	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
$sin\left(2\pi \cdot 10\frac{t}{2400}\right)$	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
$cos\left(2\pi \cdot 5\frac{t}{2400}\right)$	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
$cos\left(2\pi \cdot 15\frac{t}{2400}\right)$	1.00	1.00	0.00	1.00	1.00
$sin\left(2\pi \cdot 20\frac{t}{2400}\right)$	1.00	0.00	0.00	0.80	0.00
$sin\left(2\pi \cdot 15\frac{t}{2400}\right)$	0.35	0.00	0.00	0.35	0.00
$cos\left(2\pi \cdot 25\frac{t}{2400}\right)$	0.35	0.00	0.00	0.20	0.00
$sin\left(2\pi \cdot 3\frac{t}{2400}\right)$	0.15	0.00	0.00	0.00	0.00
$sin\left(2\pi \cdot 45\frac{t}{2400}\right)$	0.05	0.00	0.00	0.00	0.00
$cos\left(2\pi \cdot 10\frac{t}{2400}\right)$	0.05	0.00	0.00	0.00	0.00
$cos\left(2\pi \cdot 11\frac{t}{2400}\right)$	0.00	0.00	0.00	0.05	0.00

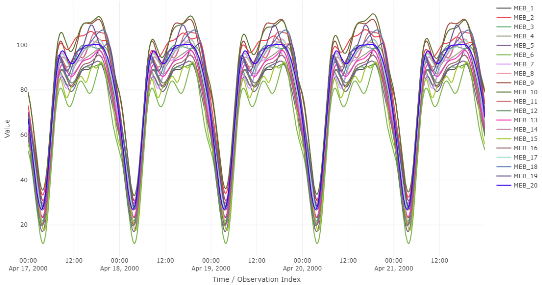
Table: *Variable Inclusion Probabilities for location L102*

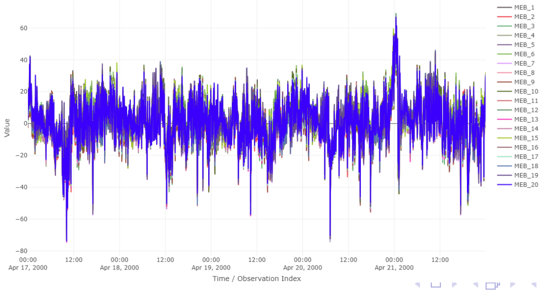
Results: L102 - Coefficient Estimates with CIs

Frequencies	**L1MTE$_{\text{AIC/BIC}}$**	**L1LSA$_{\text{AIC}}$**	**L1LSA$_{\text{BIC}}$**	**L2MTE$_{\text{AIC/BIC}}$**	**L2LSA$_{\text{AIC/BIC}}$**
(Intercept)	77.53 (66.89, 78.21 , 87.45)	76.42 (66.48, 76.37 , 86.29)	75.16 (65.07, 75.14 , 85.12)	77.82 (68.47, 77.90 , 87.35)	76.40 (66.47, 76.40 , 86.30)
$cos\left(2\pi \cdot 15\frac{t}{2400}\right)$	7.03 (5.65, 6.92 , 8.94)	5.84 (5.65, 5.82 , 6.13)	-	7.28 (5.49, 6.90 , 10.31)	6.22 (6.07, 6.20 , 6.45)
$sin\left(2\pi \cdot 20\frac{t}{2400}\right)$	4.16 (2.52, 3.84 , 6.77)	-	-	2.95 (0.00, 3.52 , 4.86)	-
$sin\left(2\pi \cdot 3\frac{t}{2400}\right)$	0.17 (0.00, 0.00 , 1.68)	-	-	-	-
$cos\left(2\pi \cdot 10\frac{t}{2400}\right)$	0.09 (0.00, 0.00 , 0.93)	-	-	-	-
$sin\left(2\pi \cdot 45\frac{t}{2400}\right)$	0.07 (0.00, 0.00 , 0.72)	-	-	-	-
$cos\left(2\pi \cdot 11\frac{t}{2400}\right)$	0.00 (-0.00, 0.00 , 0.00)	-	-	0.12 (0.00, 0.00 , 1.28)	-
$sin\left(2\pi \cdot 15\frac{t}{2400}\right)$	−0.87 (-3.59, 0.00 , 0.00)	-	-	−0.91 (-4.30, 0.00 , 0.00)	-
$cos\left(2\pi \cdot 25\frac{t}{2400}\right)$	−1.05 (-4.10, 0.00 , 0.00)	-	-	−0.51 (-3.44, 0.00 , 0.00)	-
$sin\left(2\pi \cdot 10\frac{t}{2400}\right)$	−14.71 (-16.25, -14.52 , -13.37)	−11.97 (-12.30, -11.93 , -11.78)	−8.39 (-8.69, -8.34 , -8.20)	−15.24 (-17.21, -15.03 , -14.02)	−12.23 (-12.46, -12.21 , -12.04)
$cos\left(2\pi \cdot 5\frac{t}{2400}\right)$	−17.29 (-18.94, -17.62 , -14.71)	−14.44 (-14.64, -14.44 , -14.13)	−11.36 (-11.62, -11.41 , -10.98)	−17.37 (-19.36, -17.88 , -13.80)	−14.35 (-14.58, -14.33 , -14.14)
$sin\left(2\pi \cdot 5\frac{t}{2400}\right)$	−22.93 (-25.38, -22.35 , -21.06)	−19.36 (-19.71, -19.32 , -19.10)	−16.99 (-17.38, -16.99 , -16.66)	−22.80 (-24.76, -22.54 , -20.62)	−19.10 (-19.42, -19.13 , -18.88)

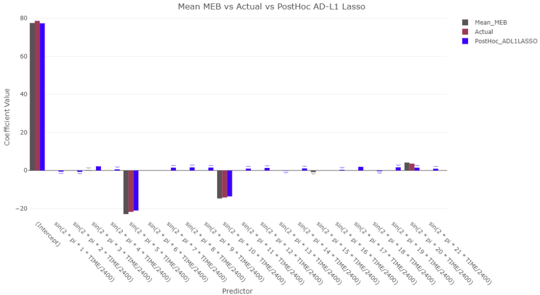
Table: *Location L102. The coral tiny numbers indicate the 2.5th and 97.5th quantiles, the navy-blue tiny numbers represent the median values, and normal-sized numbers refer to the mean value of the estimated coefficients model-wise.*

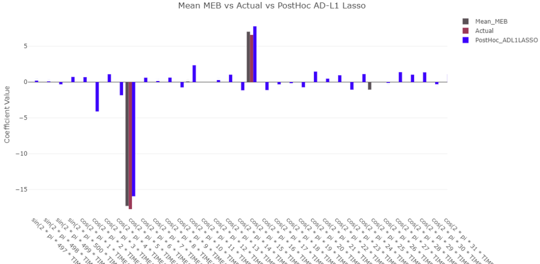
ℓ_1MTE: Forecast Results (L102



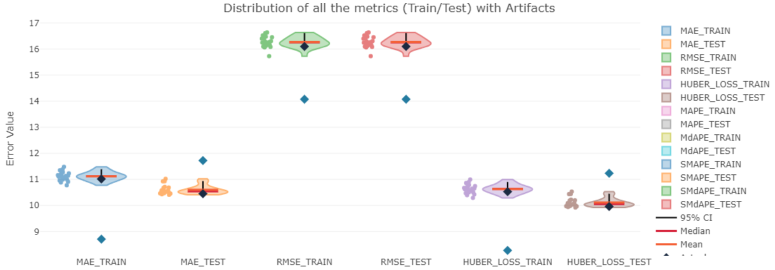


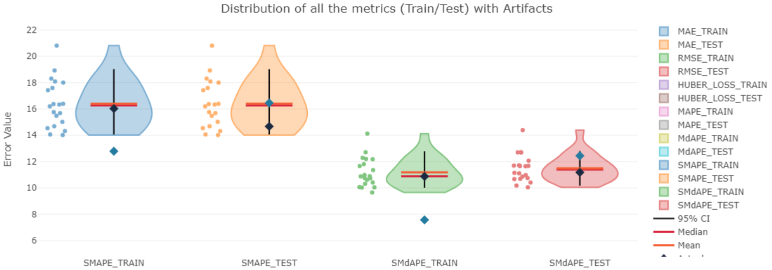
ℓ_1MTE: Coefficients ADL1LASSO vs Actual vs Model Average (L102)



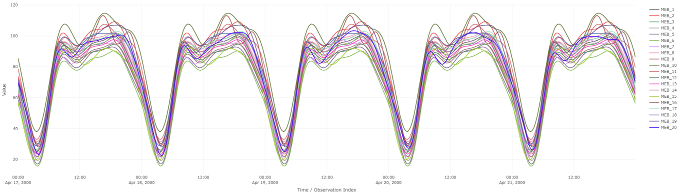


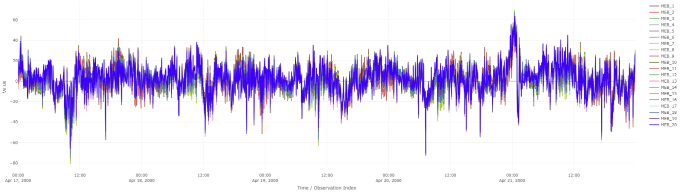
ℓ_1MTE - Violin plots of Key Performance Metrics (L102)



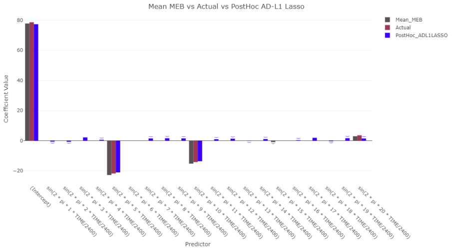


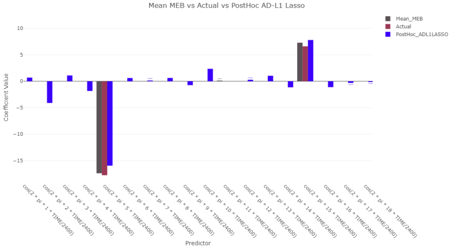
ℓ_2MTE: Forecast Results (L102)



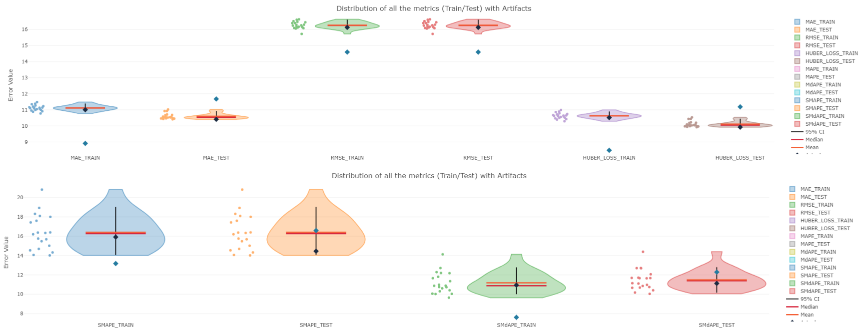


ℓ_2MTE: Coefficients ADL1LASSO vs Actual vs Model Average (L102)

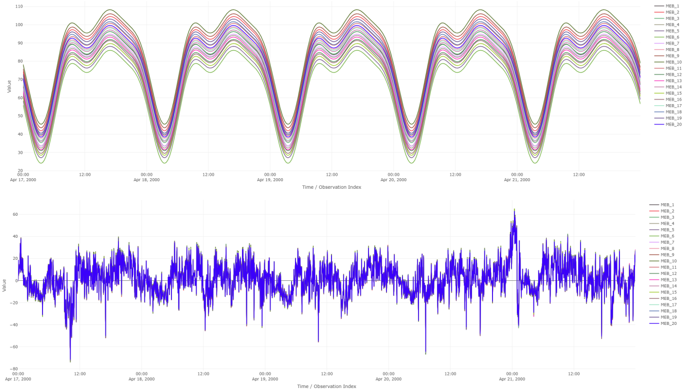




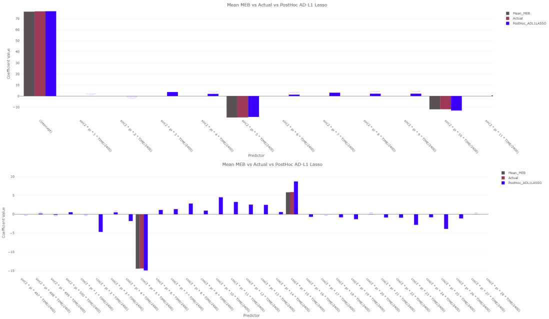
ℓ_2MTE - Violin plots of Key Performance Metrics (L102)



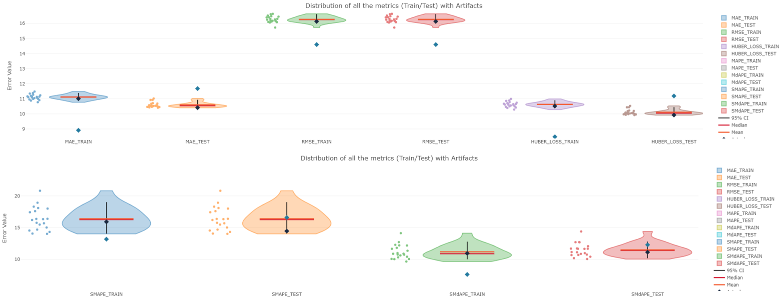
ℓ_1LSA: Forecast Results (L102)



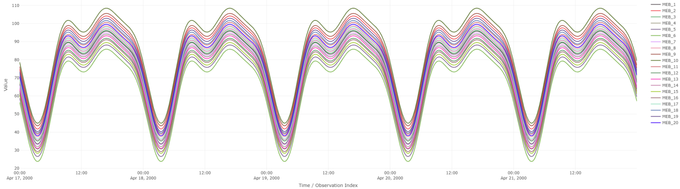
ℓ_1LSA: Coefficients ADL1LASSO vs Actual vs Model Average (L102)

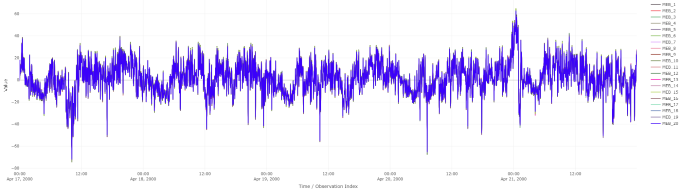


ℓ_1LSA - Violin plots of Key Performance Metrics (L102)

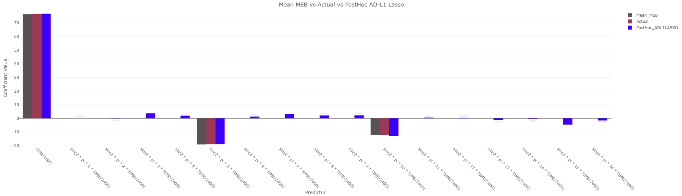


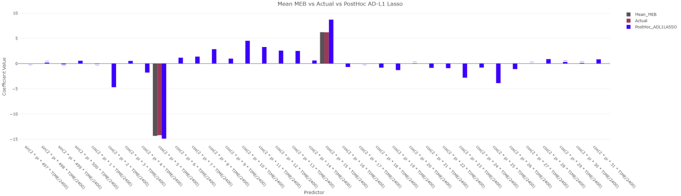
ℓ_2LSA: Forecast Results (L102)



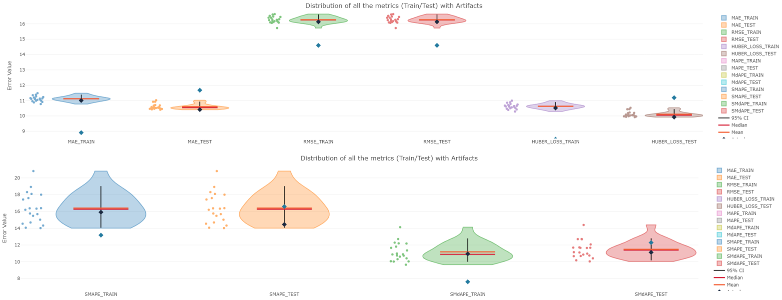


ℓ_2LSA: Coefficients ADL1LASSO vs Actual vs Model Average (L102)





ℓ_2LSA - Violin plots of Key Performance Metrics (L102)



Results: L102 - Model Forecasting

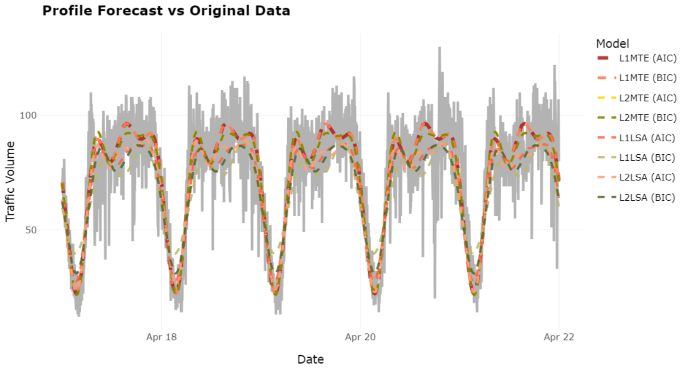


Figure: *Forecasting results of each method plotted on the actual data of location L102. The curved line with firebrick is the AIC-type of the $\ell_1 MTE$, the coral is the BIC-type alternative, the gold and the olive is for $\ell_2 MTE$ with AIC and BIC minimization ,tomato and dark khaki corresponds to for $\ell_1 LSA$, and the last couple of for $\ell_2 LSA$ light salmon and dark olive green respectfully.*

Results: Robust MTE (ℓ_1MTE) - VIPs on L102

- **ℓ_1MTE:**
  - **100% VIP:** Intercept, $\sin(2\pi \cdot xt/T)$ ($x \in \{5, 10, 20\}$), $\cos(2\pi \cdot xt/T)$ ($x \in \{5, 15\}$) under any Information criterion minimization.
  - Lower VIPs for other regressors (e.g., $\sin(2\pi \cdot 15t/T)$ (35%), $\cos(2\pi \cdot 25t/T)$ (35%)).
- **Sparsity:** Leads to highly sparse coefficient vectors.
- **Consistency:** Coefficient estimates across MEBoot samples show similar values, verifying significance.
- **Variability:** Features like $\cos(2\pi \cdot 25t/T)$ and $\sin(2\pi \cdot 15t/T)$ show more fluctuations, indicating less crucial impact.

Results: Robust MTE (ℓ_1MTE) - Performance (L102)

- **Test Phase** Model preserves predictive structure, robust generalization. Residuals remain centered around zero, no strong discernible seasonality.
- **Robustness:** Evident during periods with external factors (political campaigns, sporting events), yielding stationary-in-mean residuals.
- **Adaptive ℓ_1-Lasso:** Yields sparser estimations than Adaptive ℓ_1-Lasso, but consistent sign and relative magnitude for influential terms.
- **Error Metrics:** Model averaging of ℓ_1MTE consistently yields second-lowest Huber loss, competitive SMdAPE. Direct application achieves lowest RMSE.

Results: Conventional MTE (ℓ_2MTE) & LSA (L102)

- ℓ_2**MTE :**
  - AIC and BIC yield identical sets of significant regressors.
  - Intercept, $\sin(2\pi \cdot xt/T)$ $(x \in \{5, 10\})$, $\cos(2\pi \cdot yt/T)$ $(y \in \{5, 15\})$ are 100% VIP.
  - Coefficient estimates generally similar to ℓ_1MTE.
  - Achieves lowest HL, MAE, RMSE in testing phases under model averaging.
- ℓ_1**LSA :**
  - More pronounced shrinkage, retaining only 5 features (Intercept, $\sin(2\pi \cdot xt/T)$ $(x \in \{5, 10\})$, $\cos(2\pi \cdot xt/T)$ $(x \in \{5, 15\})$).
  - Coefficients consistent, but omission of features like $\sin(2\pi \cdot 20t/T)$ due to over-shrinkage.
  - Fitted values show strong periodicity, but residuals indicate heteroskedasticity and persistent singularities.
  - Stable but non-optimal performance in error metrics.
- ℓ_2**LSA:**
  - Consistent parameter estimation, but persistent over-shrinkage.
  - Residuals show clear seasonal pattern, indicating omitted periodic components.

Results: Comparison Across Locations (VIPs Heatmap)

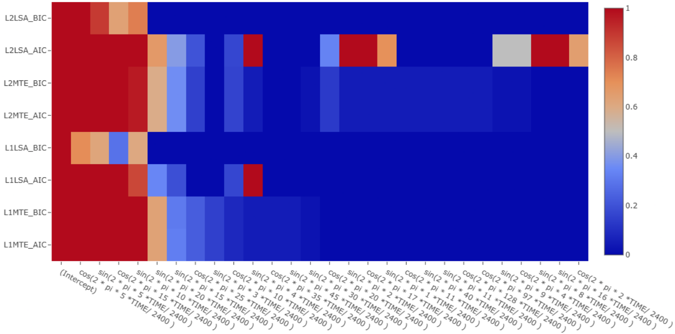


Figure: *Variable Inclusion Probabilities of each model over all the locations*

Results: Comparison Across Locations

Features \ Methods	L1MTE$_{AIC/BIC}$	L1LSA$_{AIC}$	L1LSA$_{BIC}$	L2MTE$_{AIC/BIC}$	L2LSA$_{AIC}$	L2LSA$_{BIC}$	Overall
$(Intercept)$	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
$cos\left(2\pi \cdot 5\frac{t}{2400}\right)$	**1.00**	**1.00**	**0.71**	**1.00**	**1.00**	**1.00**	**0.96**
$sin\left(2\pi \cdot 5\frac{t}{2400}\right)$	**1.00**	**1.00**	**0.62**	**1.00**	**1.00**	**0.89**	**0.94**
$cos\left(2\pi \cdot 15\frac{t}{2400}\right)$	**1.00**	**1.00**	**0.29**	**1.00**	**1.00**	**0.64**	**0.87**
$sin\left(2\pi \cdot 10\frac{t}{2400}\right)$	**1.00**	**0.86**	**0.61**	**0.96**	**1.00**	**0.74**	**0.89**
$sin\left(2\pi \cdot 20\frac{t}{2400}\right)$	0.63	0.34	0.00	0.59	0.67	0.00	0.43
$sin\left(2\pi \cdot 15\frac{t}{2400}\right)$	0.31	0.19	0.00	0.36	0.40	0.00	0.24
$cos\left(2\pi \cdot 25\frac{t}{2400}\right)$	0.23	0.00	0.00	0.15	0.20	0.00	0.12
$sin\left(2\pi \cdot 3\frac{t}{2400}\right)$	0.15	0.00	0.00	0.00	0.00	0.00	0.04
$cos\left(2\pi \cdot 10\frac{t}{2400}\right)$	0.08	0.17	0.00	0.16	0.17	0.00	0.10
$sin\left(2\pi \cdot 4\frac{t}{2400}\right)$	0.05	1.00	0.00	0.05	1.00	0.00	0.28
$cos\left(2\pi \cdot 35\frac{t}{2400}\right)$	0.05	0.00	0.00	0.00	0.00	0.00	0.01
$sin\left(2\pi \cdot 45\frac{t}{2400}\right)$	0.05	0.00	0.00	0.00	0.00	0.00	0.01
$sin\left(2\pi \cdot 30\frac{t}{2400}\right)$	0.02	0.00	0.00	0.02	0.00	0.00	0.01
$cos\left(2\pi \cdot 20\frac{t}{2400}\right)$	0.00	0.00	0.00	0.13	0.33	0.00	0.07
$sin\left(2\pi \cdot 2\frac{t}{2400}\right)$	0.00	0.00	0.00	0.05	1.00	0.00	0.14
$cos\left(2\pi \cdot 17\frac{t}{2400}\right)$	0.00	0.00	0.00	0.05	1.00	0.00	0.14
$sin\left(2\pi \cdot 1\frac{t}{2400}\right)$	0.00	0.00	0.00	0.05	0.70	0.00	0.10
$cos\left(2\pi \cdot 11\frac{t}{2400}\right)$	0.00	0.00	0.00	0.05	0.00	0.00	0.01
$sin\left(2\pi \cdot 40\frac{t}{2400}\right)$	0.00	0.00	0.00	0.05	0.00	0.00	0.01
$sin\left(2\pi \cdot 11\frac{t}{2400}\right)$	0.00	0.00	0.00	0.05	0.00	0.00	0.01
$cos\left(2\pi \cdot 128\frac{t}{2400}\right)$	0.00	0.00	0.00	0.05	0.00	0.00	0.01
$cos\left(2\pi \cdot 97\frac{t}{2400}\right)$	0.00	0.00	0.00	0.05	0.00	0.00	0.01
$sin\left(2\pi \cdot 9\frac{t}{2400}\right)$	0.00	0.00	0.00	0.02	0.50	0.00	0.07
$cos\left(2\pi \cdot 4\frac{t}{2400}\right)$	0.00	0.00	0.00	0.02	0.50	0.00	0.07
$sin\left(2\pi \cdot 8\frac{t}{2400}\right)$	0.00	0.00	0.00	0.00	1.00	0.00	0.12
$sin\left(2\pi \cdot 16\frac{t}{2400}\right)$	0.00	0.00	0.00	0.00	1.00	0.00	0.12
$cos\left(2\pi \cdot 2\frac{t}{2400}\right)$	0.00	0.00	0.00	0.00	0.65	0.00	0.08

Table: *Variable Inclusion Probabilities for according to all locations*

Results: Comparison Across Locations (Forecast VS Initial data)

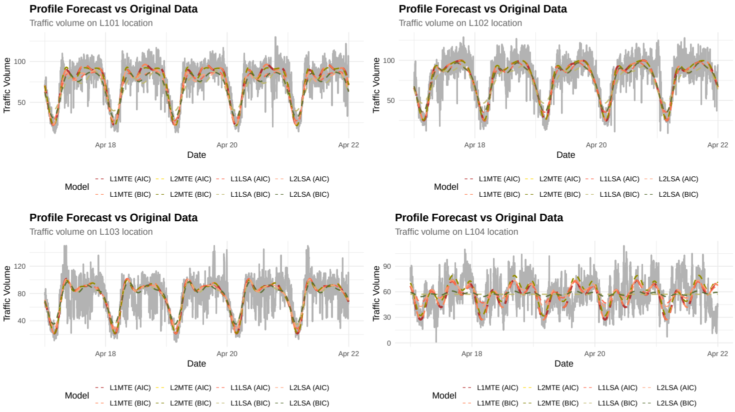


Figure: *Locationwise plots of forecasting results of each method plotted on the actual data. The curved line with firebrick is the AIC-type of the ℓ_1 MTE, the coral is the BIC-type alternative, the gold and the olive is for ℓ_2 MTE with AIC and BIC minimization ,tomato and dark khaki corresponds to for ℓ_1 LSA, and the last couple of for ℓ_2 LSA light salmon and dark olive green respectfully.*

Results: Comparison Across Locations (Forecast VS Initial data, Cont'd)

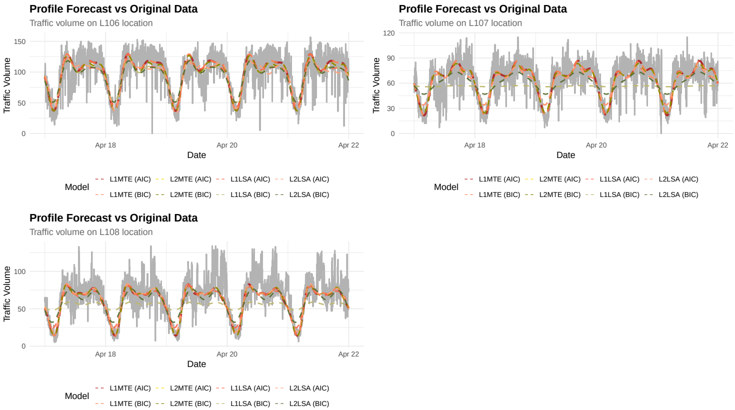


Figure: *Locationwise plots of forecasting results of each method plotted on the actual data. The curved line with firebrick is the AIC-type of the ℓ_1 MTE, the coral is the BIC-type alternative, the gold and the olive is for ℓ_2 MTE with AIC and BIC minimization ,tomato and dark khaki corresponds to for ℓ_1 LSA, and the last couple of for ℓ_2 LSA light salmon and dark olive green respectfully.*

Results: Comparison Across Locations (MAE)

Methods / Locations	Metric: Metric: Mean Absolute Error (MAE)																	
	ℓ_1**LSA**$_{\mathbf{AIC}}$			ℓ_2**LSA**$_{\mathbf{AIC}}$			ℓ_1**MTE**$_{\mathbf{AIC/BIC}}$			ℓ_2**MTE**$_{\mathbf{AIC/BIC}}$			ℓ_1**LSA**$_{\mathbf{BIC}}$			ℓ_2**LSA**$_{\mathbf{BIC}}$		
	MEB	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**
L101	10.40	10.44	11.24	10.75	10.73	12.47	**9.26**	9.53	11.51	9.37	9.71	12.37	11.38	11.40	12.48	13.31	13.26	12.48
L102	10.97	11.00	12.18	10.99	10.98	12.18	**10.28**	10.45	11.72	10.30	10.41	11.68	10.97	11.00	12.18	13.07	13.07	12.20
L103	14.26	14.31	16.49	14.32	14.36	16.49	**13.98**	14.00	16.16	14.11	14.36	16.86	14.83	14.90	16.54	14.87	14.94	16.54
L104	13.91	13.96	14.30	14.23	14.30	14.88	13.78	13.97	14.71	13.81	13.70	14.45	14.82	15.31	14.22	15.29	15.31	**13.67**
L106	15.93	15.93	18.10	15.91	15.88	19.43	15.05	15.35	18.94	14.68	**14.59**	20.14	17.30	17.30	20.09	16.88	16.88	20.09
L107	11.94	11.97	11.74	12.01	12.04	11.74	**10.74**	11.16	12.26	**10.74**	10.82	12.59	14.27	15.01	12.97	19.55	19.74	13.46
L108	11.82	11.87	12.53	12.87	12.94	12.83	**11.73**	11.80	12.46	11.85	11.80	12.53	14.43	14.46	12.96	21.79	24.57	13.53

Table: *Comparison of all the methods ($\ell_1 MTE$,$\ell_2 MTE$, $\ell_1 LSA$, $\ell_2 LSA$ under AIC and BIC minimization) under Mean Absolute Error (MAE) metric , evaluated using model averaging from MEBoot samples (`MEB`), the corresponding models applied on the original traffic data (`Actual`), and post-hoc variable selection algorithm `ADL1LASSO` models under AIC and BIC minimization criteria, with bold values indicating the best-performing metrics within each model. The $\ell_1 MTE$ and the $\ell_2 MTE$ have identical results.*

Results: Comparison Across Locations (RMSE)

Locations \ Methods	ℓ_1LSA$_{AIC}$			ℓ_2LSA$_{AIC}$			ℓ_1MTE$_{AIC/BIC}$			ℓ_2MTE$_{AIC/BIC}$			ℓ_1LSA$_{BIC}$			ℓ_2LSA$_{BIC}$		
	Metric: Root Mean Square Error (RMSE)																	
	MEB	Actual	ADL1LASSO	MEB	Actual	ADL1LASSO	MEB	Actual	ADL1LASSO	MEB	Actual	ADL1LASSO	MEB	Actual	ADL1LASSO	MEB	Actual	ADL1LASSO
L101	13.22	17.77	15.98	13.56	18.30	15.96	**12.34**	18.66	16.17	12.45	18.26	16.10	14.25	18.77	15.41	16.16	20.10	15.52
L102	14.21	16.17	**13.57**	14.22	16.19	**13.57**	13.72	16.10	14.07	13.75	16.13	14.60	14.21	16.17	**13.57**	16.24	17.94	13.62
L103	19.24	16.71	14.35	19.35	16.68	14.35	19.27	16.86	14.76	19.45	16.85	14.44	19.49	17.43	**13.97**	19.57	17.38	**13.97**
L104	17.16	17.71	**14.17**	17.36	18.21	14.68	17.59	17.91	14.76	17.60	17.84	14.48	17.97	20.59	14.66	18.55	20.58	17.56
L106	20.53	23.20	21.30	20.31	24.34	20.06	19.98	25.60	22.54	**19.85**	25.55	22.71	21.48	25.38	19.93	21.13	25.25	19.93
L107	15.25	15.96	**12.96**	15.31	15.96	**12.96**	14.45	15.83	14.02	14.63	15.93	14.21	17.45	19.07	14.17	22.61	23.37	14.60
L108	16.61	15.02	12.51	17.56	15.70	**12.45**	16.73	15.08	12.72	16.88	15.08	12.51	18.99	16.98	12.53	26.28	26.00	13.03

Table: *Comparison of all the methods ($\ell_1 MTE$,$\ell_2 MTE$, $\ell_1 LSA$, $\ell_2 LSA$ under AIC and BIC minimization) under Root Mean Square Error (RMSE) metric , evaluated using model averaging from MEBoot samples (`MEB`), the corresponding models applied on the original traffic data (`Actual`), and post-hoc variable selection algorithm `ADL1LASSO` models under AIC and BIC minimization criteria, with bold values indicating the best-performing metrics within each model. The $\ell_1 MTE$ and the $\ell_2 MTE$ have identical results.*

Methods / Locations	Metric: Metric: Huber Loss (HL)																	
	ℓ_1**LSA**$_{\mathbf{AIC}}$			ℓ_2**LSA**$_{\mathbf{AIC}}$			ℓ_1**MTE**$_{\mathbf{AIC/BIC}}$			ℓ_2**MTE**$_{\mathbf{AIC/BIC}}$			ℓ_1**LSA**$_{\mathbf{BIC}}$			ℓ_2**LSA**$_{\mathbf{BIC}}$		
	MEB	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**
L101	9.91	9.95	10.76	10.26	10.24	11.98	**8.77**	9.04	11.02	8.89	9.22	11.88	10.89	10.91	11.99	12.81	12.77	11.99
L102	10.48	10.51	11.69	10.50	10.49	11.69	**9.79**	9.96	11.23	9.81	9.93	11.19	10.48	10.51	11.69	12.58	12.58	11.71
L103	13.77	13.82	15.99	13.83	13.87	15.99	**13.49**	13.51	15.67	13.62	13.86	16.36	14.34	14.41	16.05	14.38	14.44	16.05
L104	13.41	13.47	13.81	13.74	13.81	14.39	13.29	13.48	14.22	13.32	13.21	13.96	14.33	14.81	13.73	14.80	14.81	**13.18**
L106	15.43	15.43	17.61	15.41	15.38	18.93	14.56	14.86	18.45	14.18	**14.10**	19.64	16.80	16.80	19.60	16.38	16.39	19.60
L107	11.45	11.48	11.25	11.52	11.55	11.25	**10.26**	10.67	11.77	**10.26**	10.33	12.10	13.78	14.52	12.48	19.06	19.24	12.97
L108	11.34	11.38	12.04	12.38	12.45	12.34	**11.24**	11.31	11.97	11.36	11.31	12.04	13.94	13.97	12.47	21.29	24.07	13.04

Table: *Comparison of all the methods ($\ell_1 MTE$,$\ell_2 MTE$, $\ell_1 LSA$, $\ell_2 LSA$ under AIC and BIC minimization) under Huber Loss metric , evaluated using model averaging from MEBoot samples (`MEB`), the corresponding models applied on the original traffic data (`Actual`), and post-hoc variable selection algorithm `ADL1LASSO` models under AIC and BIC minimization criteria, with bold values indicating the best-performing metrics within each model. The $\ell_1 MTE$ and the $\ell_2 MTE$ have identical results.*

Results: Comparison Across Locations (SMAPE)

Methods / Locations	Metric: Symmetric Mean Absolute Percentage Error (SMAPE)																	
	ℓ_1**LSA**$_{\text{AIC}}$			ℓ_2**LSA**$_{\text{AIC}}$			ℓ_1**MTE**$_{\text{AIC/BIC}}$			ℓ_2**MTE**$_{\text{AIC/BIC}}$			ℓ_1**LSA**$_{\text{BIC}}$			ℓ_2**LSA**$_{\text{BIC}}$		
	MEB	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**
L101	15.49	15.63	16.79	16.34	16.40	19.34	**13.87**	14.14	17.68	14.14	14.98	19.57	17.29	17.41	18.78	20.81	20.82	18.84
L102	15.96	16.05	17.16	16.02	16.07	17.16	14.36	14.67	16.46	**14.31**	14.46	16.59	15.96	16.05	17.16	19.75	19.80	17.24
L103	20.66	20.82	22.99	20.67	20.79	22.99	**19.71**	19.76	22.40	19.88	20.13	23.36	22.20	22.37	23.11	22.19	22.37	23.11
L104	25.27	25.39	25.86	26.01	26.14	27.29	24.54	24.79	25.82	25.11	**24.45**	26.44	27.18	27.98	25.06	27.95	27.98	24.60
L106	17.51	17.55	19.97	17.78	17.79	21.64	16.69	17.05	20.97	16.38	**16.37**	22.77	19.78	19.83	22.56	19.31	19.37	22.56
L107	21.71	21.82	20.24	21.82	21.92	20.24	18.73	19.87	21.15	**18.41**	18.50	21.90	26.34	27.61	22.48	34.84	35.13	23.21
L108	19.41	19.64	19.96	22.05	22.33	20.72	**18.77**	18.86	19.86	18.94	18.87	19.96	25.62	25.77	21.02	38.20	42.87	22.34

Table: *Comparison of all the methods ($\ell_1 MTE$,$\ell_2 MTE$, $\ell_1 LSA$, $\ell_2 LSA$ under AIC and BIC minimization) under Symmetric Mean Absolute Percentage Error (SMAPE) metric , evaluated using model averaging from MEBoot samples (`MEB`), the corresponding models applied on the original traffic data (`Actual`), and post-hoc variable selection algorithm `ADL1LASSO` models under AIC and BIC minimization criteria, with bold values indicating the best-performing metrics within each model. The $\ell_1 MTE$ and the $\ell_2 MTE$ have identical results.*

Results: Comparison Across Locations (SMdAPE)

Methods / Locations	Metric: Symmetric Median Absolute Percentage Error (SMdAPE)																	
	ℓ_1**LSA**$_{\text{AIC}}$			ℓ_2**LSA**$_{\text{AIC}}$			ℓ_1**MTE**$_{\text{AIC/BIC}}$			ℓ_2**MTE**$_{\text{AIC/BIC}}$			ℓ_1**LSA**$_{\text{BIC}}$			ℓ_2**LSA**$_{\text{BIC}}$		
	MEB	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**	**MEB**	**Actual**	**ADL1LASSO**
L101	12.58	12.68	12.11	12.71	12.57	13.27	**10.23**	10.33	12.12	10.38	10.75	13.31	14.02	13.94	12.76	16.48	16.29	12.58
L102	11.94	11.98	12.62	11.86	11.86	12.62	**10.96**	11.18	12.44	11.04	11.12	12.29	11.94	11.98	12.62	14.40	14.34	12.55
L103	14.24	14.21	16.66	14.27	14.32	16.66	13.96	13.96	16.26	14.10	**13.95**	17.17	15.00	14.96	16.72	15.09	15.12	16.72
L104	20.77	20.88	**19.07**	21.83	22.01	19.61	19.92	20.02	19.81	21.19	19.28	19.37	23.49	23.97	19.07	23.71	23.86	19.87
L106	13.87	13.90	14.19	13.99	13.89	15.18	12.21	12.26	14.75	**11.50**	11.54	15.16	15.46	15.39	15.73	14.81	14.73	15.73
L107	16.06	16.01	14.86	16.29	16.25	14.86	13.46	13.96	15.41	**13.12**	13.22	15.74	19.42	20.13	16.48	30.52	30.69	16.54
L108	14.56	14.62	15.56	16.49	16.59	15.48	**14.35**	14.52	15.43	14.67	14.38	15.56	19.01	18.96	15.48	31.18	36.40	15.89

Table: *Comparison of all the methods ($\ell_1 MTE$,$\ell_2 MTE$, $\ell_1 LSA$, $\ell_2 LSA$ under AIC and BIC minimization) under Symmetric Median Absolute Percentage Error (SMdAPE) metric , evaluated using model averaging from MEBoot samples (`MEB`), the corresponding models applied on the original traffic data (`Actual`), and post-hoc variable selection algorithm `ADL1LASSO` models under AIC and BIC minimization criteria, with bold values indicating the best-performing metrics within each model. The $\ell_1 MTE$ and the $\ell_2 MTE$ have identical results.*

Results: Comparison Across Locations

- **Pervasive Sparsity:** Across all locations, most estimated coefficients converge to near zero.
- **Key Predictors:** Intercept, $\sin(2\pi \cdot 5t/2400)$, $\cos(2\pi \cdot 5t/2400)$, $\sin(2\pi \cdot 10t/2400)$, and $\cos(2\pi \cdot 15t/2400)$ are consistently influential.
- **AIC vs. BIC :**
  - BIC: Consistently higher sparsity (cooler colors/lower VIPs).
  - AIC: Retains larger ensemble of predictors (warmer colors/higher VIPs).
- **LSA vs. MTE Stability:**
  - LSA models: Variable selection is unstable, sensitive to information criterion.
  - MTE models: More consistent variable selection between AIC and BIC.
- **Forecasting Performance :** ℓ_1MTE, regardless of information criteria, substantially surpasses other models in capturing periodicity. LSA-based models show poor performance.

Key Contributions & Future Work

- **MTE Superiority:** MTE-based models consistently demonstrate superior performance, robustness, and stability for traffic forecasting.
  - Better predictive accuracy, robustness, and error metric minimization (e.g., average 10.28% increase in HL from MTE to LSA).
  - Stable feature inclusion regardless of AIC/BIC.
  - Better generalization to unseen data.
- **LSA Limitations:** Performance limited by over-shrinkage and naive covariance matrix estimation.
- **Traffic Flow Dynamics:** Driven by a sparse set of low-frequency periodic components (e.g., $\sin(2\pi \cdot 5t/2400)$, $\cos(2\pi \cdot 5t/2400)$).
- **Adaptive ℓ_1-LASSO:** Proposed framework, though with limited forecasting improvements due to false positives in feature selection.
- **Future Work:**
  - Adopt more robust covariance matrix estimation methods (Galvao & Yoo [Galvao and Yoon, 2021]n, Newey & West[Newey and West, 1994]).
  - Extend Taylor expansion order for MTE.
  - Significantly increase MEBoot samples (e.g., to 999).

Conclusion

- ℓ_1**MTE-based models outperform LSA counterparts**.
- The resulting time series are **stationary in mean**.
- The SMAPE and SMdAPE values indicate that **Practitioners can use MTE-based** models as a naive implementation. It could be used as benchmark or for handling missing values.
- The adaptive ℓ_1 LASSO with VIP-based weights did not perform well.
- Rodrigues's work [Rodrigues, 2023] can be extended for robust variable selection, showing promising results.

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Thank you!

THE END